For A some $n \times n$ symmetric matrix, we defined quadratic form, Q from $\mathbb{R}^n$ to $\mathbb{R}$ as

$$Q(x) = x^T A x$$

We did example where $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ which gave

$$Q(x) = x_1^2 + 4x_1x_2 + 3x_2^2$$

again this term, cross product term, is annoying.

For one, makes it harder to tell if this is always positive, always negative, or neither.

And why did we insist that A has to be symmetric? Seems like this calculation would still work perfectly fine if A were not symmetric?

To address these issues consider the following:

Say we did a sort of change of variables, linear style so no overall change in degree. But maybe can make it easier to work with, perhaps eliminate the cross product term?

To that end, say for P invertible we replace $x = Py$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \text{and as usual} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Again, linear change of variables. Then we see that

$$Q(y) = (Py)^T A(Py) = y^T(P^T AP)y$$

Suppose P^TAP is a diagonal matrix so that this quadratic form, in y, won't have any cross product terms - naturally might be easier!

Is that always possible though? Well A is symmetric so

$$A = PDP^{-1} = PDP^T$$

for some P orthogonal, D diagonal. For those we have

$$P^TAP = D$$

Hence we choose that P for $x = Py$ to guarantee this. So yes this is always possible. But you see now a big reason why in quadratic forms we insist A be symmetric even if calculation still makes sense without it being so. This diagonalization strategy simply wouldn't work if A wasn't symmetric.

For one if A is not symmetric, it might not even be diagonalizable. And even if it is, the diagonalization happening with the help of the transpose was something that was tied to symmetric matrices, definitely did not necessarily work like that with nonsymmetric matrices.

7.2.7

$$Q(x) = Q\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = x_1^2 + 10x_1x_2 + x_2^2$$

For this based on the squared coefficients and such:

$$Q = x^T A x = x^T \begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix} x$$

so that

$$A = \begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}$$

So we need to orthogonally diagonalize A:

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 5 \\ 5 & 1-\lambda \end{bmatrix}$$

----> $\det(A - \lambda I) = (1-\lambda)^2 - 25 \overset{?}{=} 0$

----> $\lambda = -4, 6$

Thus

$$A + 4I = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

so an eigenvector here is $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

And for the other eigenvalue:

$$A - 6I = \begin{bmatrix} -5 & 5 \\ 5 & -5 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$$

so an eigenvector for this is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Hence $A = PDP^{-1}$ for

$$P = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \quad , \quad D = \begin{bmatrix} -4 & 0 \\ 0 & 6 \end{bmatrix}$$

Thus using y obtained from $x = Py$ we will have

$$Q(y) = Q\left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) = -4y_1^2 + 6y_2^2$$

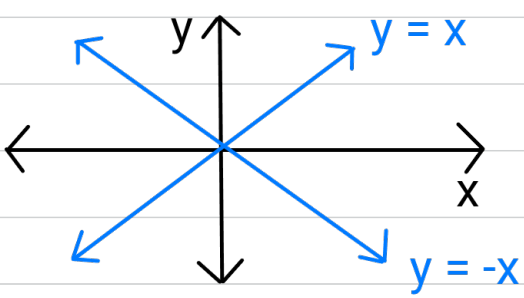
just via D as that is the symmetric matrix now used.

So again in y there is no cross product term, perhaps that's easier.

But you might ask why changing the variable alters this. After all, these variables are just names, why should it matter/change anything?

Maybe a better way to look at that which clarifies this is we are really changing the <u>principal axes</u>.

What does that mean? Well if in $\mathbb{R}^2$ say, someone makes reference to the "axes" what do they mean? Well naturally you would think they mean x and y axes. However, consider



If we instruct someone to walk 5 units on the "first axis" you would think again I meant x-axis. But what if my "axes" were the two lines shown, $y = x$ and $y = -x$?

Of course that doesn't seem natural but nothing inherently wrong with that.

In example shown, the columns of P were essentially

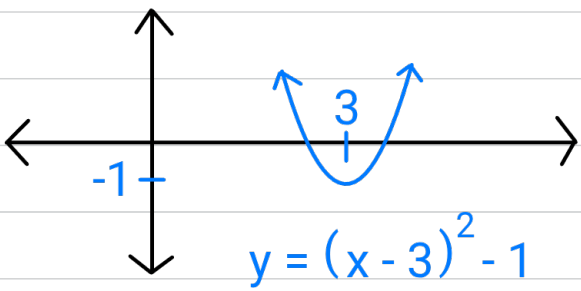
$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, that is the two lines mentioned above.

So thinking of those as the axes, i.e, the principal axes, under that orientation the quadratic form has no cross product terms is point.

This is called the <u>Principal Axis Theorem</u>.

That is, in this approach, using the columns of P as the axes, the quadratic form won't have any cross product terms so will be easier to assess its nature.

A good way to understand this better is to see:



This is a parabola whose axis of symmetry is $x = 3$, parallel to the y-axis, one of the traditional principal axes.

Hence why in the normal variables this has no cross product term.

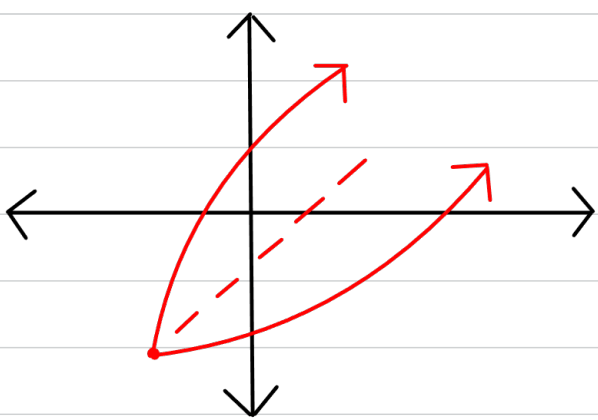
So let's say we know the axis of symmetry and a few points, could we find the equation (assuming it wasn't given)?

Probably you could as based on the axis of symmetry you would know the equation is of form:

$$y = a(x - 3)^2 + b$$

just have to extrapolate constants a and b.

However now consider instead



This parabola clearly has its axis of symmetry not a horizontal or vertical line so its equation has an xy term.

That is, it has a cross product term, obviously annoying.

So if we gave that axis plus some points could we find the equation of the parabola as we proposed to for the prior one?

Doable but much more complicated, the presence of the xy term makes it much more annoying.

What if we instead took the principal axes to be the line through the origin parallel to this axis of symmetry with the other axis being perpendicular to that (also through the origin). Key is with that change of variables, this parabola won't have a cross product term like the one like the prior form which used the x and y axes. Maybe easier to find its equation then?

So motivation is a bit clearer. Can expand on this with how equations with second degree or less terms work in general.

An interesting application of this is to consider the set of curves

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

for constants $A, B, \ldots, F$ with A, B, C not all 0. What shape is this curve?

Turns out the sign of $B^2 - 4AC$ is the key. If positive then this is a hyperbola, zero parabola, negative ellipse.

Why? If we did a change of variables to eliminate the mixed term Bxy, the associated symmetrix matrix would be:

$$\begin{bmatrix} A & \frac{B}{2} \\ \frac{B}{2} & C \end{bmatrix}$$

And determinant of this is $AC - \frac{B^2}{4}$ multiple of above. So not surprising the sign of the determinant of this matrix ends up being the key.

To clarify to this, we generalize a bit:

A quadratic form Q is <u>positive definite</u> if $Q(x) > 0 \ \forall \ x \neq 0$. Naturally <u>negative definite</u> if $Q(x) < 0 \ \forall \ x \neq 0$

When does the former hold? Based on the diagonalization ideas, exactly when the eigenvalues of the associated matrix A are all positive.

Similarly Q is <u>positive semidefinite</u> if $Q(x) \geq 0$ always, presumably being 0 at some nonzero x (maybe not required, but if not matrix would be positive definite so we'd just say that).

And this will happen when all eigenvalues of A are ≥ 0 with 0 presumably being one of the eigenvalues.

If Q can take both positive and negative values, then Q is <u>indefinite</u>. And this naturally happens when A has at least one positive eigenvalue as well as one negative.

So returning to

$$\begin{bmatrix} A & \frac{B}{2} \\ \frac{B}{2} & C \end{bmatrix}$$

which has determinant $AC - \frac{B^2}{4}$ recalling that in a square matrix of even size that determinant is product of eigenvalues, when this determinant is positive the two eigenvalues must be of the same sign (both nonzero), making the quadratic form positive or negative definite.

And that makes the curve an ellipse as $2x^2 + y^2 = 1$ logically suggests.

In contrast, if that determinant is negative then the two eigenvalues are opposite sign (both nonzero) making the quadratic form indefinite, and the curve becomes a hyperbola as $2x^2 - y^2 = 1$ suggests.

Finally if determinant is 0 so exactly one eigenvalue is 0 (both can't be 0 in symmetric matrix unless it's the 0 matrix which we are not allowing), then the quadratic form is positive or negative semidefinite, and curve is then a parabola as $-x^2 + y = 0$ suggests.

So those match the "test" of the determinant sign suggested before.

This is the end of the required material for Midterm 2.

7.4 Singular Value Decomposition (SVD)

Chapter 5 was largely about writing square matrix A in the form

$$A = PDP^{-1}$$

One thing done in Chapter 6 was for matrix with linearly independent columns A write

$$A = QR$$

where Q same size as A with orthonormal columns and R square of appropriate size which is upper triangular and has positive diagonal entries.

What was intent with these? Well you break a matrix down into much simpler matrices. But does that help? Sure the pieces are simpler, but the fact that there are multiple of them instead of one single matrix seems double edged.

Turns out it is better, especially for machines. And we saw diagonalization had other uses like making it easier to compute say A^{100}

So diagonalization and QR Factorization have several uses. Unfortunately both are restricted as diagonalization only works for certain square matrices and QR Factorization only works for matrices with linearly independent columns.

Hence why we introduce <u>Singular Value Decomposition (SVD)</u>, as this works for any size matrix, no restrictions of any kind.

Part of point is breaking data down into simpler matrices - is again better for machines, huge in data compression. And when you are using it for that type of purpose you often don't have any control of the size of the data, you can't guarantee the associated matrix will be square say. Hence we want the generalized SVD.

An example where you might use this. Not really of the size where'd you do this, but an example maybe you can appreciate from a real world standpoint.

Say thinking about building your schedule for future semester before times are announced:

Class / Time	A	B	C	D
8				
9				
10				
11				

So you are thinking of what classes to take and ranking them based on priority with the time the class also perhaps affecting its priority in your eyes. Thus you rank each class say from 1 - 10 based on importance and time.

How to analyze this data concretely to make a plan? That's what SVD is about, breaking things down into three simpler parts to analyze it more easily. With a very large table breaking things down could be quite valuable!

So the actual breakdown for A any $m \times n$ real matrix is

$$A = U \Sigma V^T$$

U: $m \times m$ orthogonal matrix
Σ: $m \times n$ "diagonal" matrix
V: $n \times n$ orthogonal matrix

Note that Σ being "diagonal" is a bit odd since diagonal usually only used for square matrices and Σ is usually not square.

But fortunately the definition is basically the same.

If as usual the ij-th entry is denoted as say b_{ij}
we must have $b_{ij} = 0 \quad \forall \; i \neq j$

So if matrix is not square, it has extra rows or columns all of which are 0 is the main issue.

In the class/time example mentioned, one of class/time functions as U, other as V with the class ranking behaving as Σ is the point.

Big question is how do we know every matrix has such a breakdown? After all diagonalization and QR Factorization didn't always work so why should this? And if we verify we can always find such a breakdown how do we actually find that breakdown?

Will answer the first question by showing the second, illustrating how it works for any matrix and that that construction always works which will retroactively show the first statement.